

Douglas Coates

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arXiv arxiv.org/a/coates_d_1

RESEARCH POSITIONS

Postdoctoral Research , FAPERJ Nota 10 position under the supervision of Katrin Gelfert, Universidade Federal do Rio de Janeiro, Brazil	April — Present
Postdoctoral Researcher , FAPESP funded position under the supervision of Ali Tahzibi, Universidade de São Paulo, São Carlos, Brazil	Jul. 2023 — Apr. 2025
Postdoctoral Research Associate , EPSRC funded position under the supervision of Mark Holland, The University of Exeter, Exeter, United Kingdom	Apr. 2022 — Apr. 2023
Postdoctoral Researcher , under the supervision of Sylvain Crovisier, Université Paris Saclay (CNRS), Paris, France	Sep. 2021 — Mar. 2022

PUBLICATIONS AND PREPRINTS

- [1] Douglas Coates and Katrin Gelfert. *Hausdorff Dimension of Sets of Generic Points for Non-statistical Dynamical Systems*. Aug. 29, 2025. arXiv: [2509.00241](https://arxiv.org/abs/2509.00241) [math].
Accepted for publication in Ergodic Theory and Dynamical Systems.
- [2] Douglas Coates and Ian Melbourne. *Global-Local Mixing for Intermittent Maps with Multiple Neutral Fixed Points*. May 12, 2025. arXiv: [2505.07492](https://arxiv.org/abs/2505.07492) [math].
- [3] Douglas Coates and Stefano Luzzatto. “Persistent Non-Statistical Dynamics in One-Dimensional Maps”. In: *Communications in Mathematical Physics* 405.4 (Apr. 9, 2024), p. 102. ISSN: 1432-0916. DOI: [10.1007/s00220-024-04957-0](https://doi.org/10.1007/s00220-024-04957-0).
- [4] Douglas Coates, Ian Melbourne, and Amin Talebi. *Natural Measures and Statistical Properties of Non-Statistical Maps with Multiple Neutral Fixed Points*. July 9, 2024. arXiv: [2407.07286](https://arxiv.org/abs/2407.07286) [math].
Accepted for publication in the Journal of the London Mathematical Society
- [5] Douglas Coates, Stefano Luzzatto, and Mubarak Muhammad. “Doubly Intermittent Full Branch Maps with Critical Points and Singularities”. In: *Communications in Mathematical Physics* 402.2 (Sept. 1, 2023), pp. 1845–1878. DOI: [10.1007/s00220-023-04766-x](https://doi.org/10.1007/s00220-023-04766-x).
- [6] Douglas Coates. “Semi-Stable Laws for Intermittent Maps”. University of Exeter, 2021. URL: <https://ore.exeter.ac.uk/repository/handle/10871/124410>.
- [7] Douglas Coates, Mark Holland, and Dalia Terhesiu. “Limit Theorems for Wobbly Interval Intermittent Maps”. In: *Studia Mathematica* 261.3 (2021), pp. 269–305. ISSN: 0039-3223. DOI: [10.4064/sm200427-21-11](https://doi.org/10.4064/sm200427-21-11).

EDUCATION

PhD. Mathematics , University of Exeter, under the supervision of Mark Holland and Dalia Terhesiu.	Oct. 2016 – Jan. 2021
MSci Mathematics , University of Bristol.	Sep. 2012 – Sep. 2016

GRANTS AND AWARDS

Heilbronn Small Grants Scheme grant to support a two-week scientific visit at University of Exeter to work with Tanja Schindler on: <i>Empirical measures for interval maps with indifferent fixed points</i> .	Feb. 2026
FAPERJ Nota 10 Postdoctoral Fellowship to support the project: <i>Multi-fractal analysis and statistical properties of infinite measure preserving systems</i> .	Apr. 2025 — Jan. 2027
FAPESP Postdoctoral Fellowship to support the project: <i>Statistical and non-statistical properties of dynamical systems</i> .	Jul. 2023 — Apr. 2025

TEACHING, SUPERVISION AND OUTREACH ACTIVITIES

Teaching positions

Lecturer , International Mathematics Master, Algeria <i>Responsibilities</i> : lecturing, course design, exam writing, <i>Courses</i> : 30 hour post-graduate course in probability theory	Dec. 2024
Teaching Assistant , Universidade de São Paulo (ICMC), São Carlos, Brazil <i>Responsibilities</i> : lecturing, <i>Courses</i> : post-graduate course in ergodic theory,	Aug. 2023 – present
Academic Program Manager and Senior Tutor , International Mathematics Master (ICTP-UNESCO development program in COMSATS University, Lahore, Pakistan), <i>Responsibilities</i> : problems classes, lecturing, course preparation, administration, exam writing, supervision of master's thesis <i>Courses</i> : post-graduate courses in commutative algebra, real analysis and measure theory, differential geometry, probability, dynamical systems, functional analysis, differential equations	Jul. 2020 – Jan. 2022
Teaching Assistant at ICTP, Trieste, Italy <i>Responsibilities</i> : problems classes, homework marking <i>Courses</i> : dynamical systems and ergodic theory (master's level)	Feb. – Aug. 2021
Teaching Assistant at the University of Exeter, Exeter, UK <i>Responsibilities</i> : problems classes, exam marking, homework marking <i>Courses</i> : mathematical structures (1st year), mathematical methods (1st year), signals series and transforms (2nd/3rd year), stochastic processes (2nd/3rd year), metric spaces and topology (3rd/4th year)	2016 – 2020

Short courses

Mixing for infinite measure systems – 4 hour mini-course, Imperial College,	Jan. 2026
Some statistical properties of non-statistical maps – 6 hour mini-course, ICMC Universidade de São Paulo,	May 2024
Markov-chains and the Perron-Frobenius theorem – 6 hour online preparatory mini-course for the “VII Escola Brasileira de Sistemas Dinâmicos”	Oct. 2023
Limit theorems in dynamical systems using transfer operator methods – 6 hour mini-course joint between PUC-UFRJ	Jun. 2023

VISITING POSITIONS

Visitor of <i>Katrin Gelfert</i> , Universidade Federal do Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil	Jun. 2023
Visitor of <i>Lorenzo Diaz</i> , Pontifícia Universidade Católica do Rio de Janeiro (PUC), Rio de Janeiro, Brazil	May 2023
Visitor of <i>Ali Tahzibi</i> , University of São Paulo (USP), ICMC, São Carlos, Brazil	Nov. 2022
Visitor of <i>Marcelo Viana</i> , Instituto de Matemática Pura e Aplicada (PUC), Rio de Janeiro, Brazil	Oct. – Nov. 2022
Visitor of <i>Stefano Luzzatto</i> , The Abdus Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy	May – Dec. 2022
Visitor of <i>Stefano Luzzatto</i> , The Abdus Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy	Feb. – Aug. 2021

SELECTED RESEARCH TALKS

<i>Global-local mixing for intermittent maps</i> , DinAmici Seminar	Mar. 2026
<i>Almost-sure behaviour of empirical measures for intermittent maps</i> , RDS Rio 2026 – International Conference on Random Dynamical Systems and Related Fields,	Mar. 2026
<i>Evolution of measures for systems with parabolic fixed points</i> , University of Exeter,	Feb. 2026
<i>Dimensions of generic points for non-statistical interval maps</i> , University of Bristol,	Feb. 2026
<i>Distributional convergence of empirical measures and strong natural measures for non-statistical systems.</i> , Imperial College,	Jan. 2026
<i>Hausdorff dimension of sets of generic points for non-statistical interval maps</i> , Universidade do São Paulo,	Aug. 2025
<i>Global-local mixing for maps with several neutral fixed points</i> , Brazilian Colloquium of Mathematics, IMPA Brazil	Jul. 2025
<i>Weak physical measures and statistical properties of some non– statistical maps</i> , Dynamical systems seminar Universidade Federal do Rio de Janeiro, Brazil,	Apr. 2024
<i>Persistent non-statistical behaviour in one-dimension</i> , Conference on Dynamics and Finance: from KAM Tori to ETFs, International Centre for Theoretical Physics (ICTP), Trieste, Italy	Oct. 2023
<i>Persistent non-statistical behaviour in one-dimension</i> Dynamical systems seminar Instituto de Matematica pura e Aplicada, Rio de Janeiro, Brazil	July 2023
<i>Persistent non-statistical behaviour for interval maps with neutral fixed points</i> EDA seminar, Pontifícia Universidade Católica do Rio de Janeiro, Brazil	May 2023
<i>Persistent non-statistical behaviour for interval maps with neutral fixed points</i> University of Exeter, Exeter, United Kingdom	Apr. 2023
<i>Limit theorems for intermittent maps with critical points</i> Dynamical systems seminar, Universidade de São Paulo, São Carlos, Brazil	Nov. 2022
<i>Intermittency and non-statistical behaviour</i> Jangada Dinamica Conference – Universidade Federal do Ceará Fortaleza, Brazil	Nov. 2022

Stable laws for intermittent maps with critical points and/or singularities Dynamical systems seminar, Université Paris Saclay, Paris, France

Sep. 2021

EVENTS ORGANISED

[Probabilistic methods for non-stationary dynamical systems](#) - two day meeting and workshop on the applications of probabilistic techniques to random and non-stationary dynamical systems. *University of Exeter*

April 2023

LANGUAGES

English – Fluent,

Italian – Fluent,

Spanish – Fluent,

Portuguese – Profficient.